

Soft Sensors Enable Real-Time Ion Concentration Measurements

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Abstract

Analytical techniques such as Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES) are widely used to measure ion concentrations, but are time-consuming, labor-intensive, and costly. The required sample collection, preparation, and offline, energy-intensive analysis render ICP-OES inadequate for real-time monitoring in membrane processes. Conductivity sensors provide rapid, in situ measurements of the electrical conductivity of ionic solutions, offering an attractive alternative for real-time tracking of ionic species during membrane processes; however, converting conductivity measurements into ion concentrations in multicomponent systems remains a significant challenge. This work addresses this limitation by developing soft sensors that estimate ion concentrations in real time using conductivity data from inexpensive sensors and mathematical models that map conductivity measurements to ion concentrations. Specifically, the Shedlovsky and mean spherical approximation (MSA) models are deployed, and their conductivity predictions are validated using experimental data representative of electronic waste and lithium-ion battery leachate. By capturing key physical phenomena in ion transport in the presence of the electric field generated by the conductivity sensor, such as electrophoretic and relaxation effects, these models accurately predict the conductivity of single, binary, and ternary salt feed solutions. Furthermore, Shedlovsky and MSA-based soft sensors are applied to estimate real-time ion concentrations in the permeate solution of a membrane-based separation, and the estimates are validated against ICP-OES analysis, demonstrating good agreement (within 5%). Overall, this study demonstrates that the development of scientifically rigorous and generalizable soft sensors is essential for enabling rapid, reliable, and efficient measurement of ion concentrations in complex ionic environments.

Keywords: Conductivity, mean spherical approximation, nanofiltration membrane, critical minerals, Inductively Coupled Plasma Optical Emission Spectroscopy

1. Introduction

Direct measurement of ion concentrations in ionic solutions is inherently challenging and expensive because ions are extremely small (on the order of 1 Å), are typically solvated by surrounding solvent molecules and counter-ions, and often exist at dilute concentrations. Analytical techniques, such as Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES) are commonly used to measure the concentration of cations in ionic solutions, including membrane feed and product streams [1–3]. Despite their accuracy, these analytical techniques are time-consuming, labor intensive, and often require more than an hour for sample collection, preparation, and analysis [1]. Consequently, such methods are inadequate for rapidly tracking ion concentrations in continuous systems where the process conditions evolve with time. Reliance on these offline techniques for dynamic systems may, therefore, lead to delayed or misleading representation of the real-time physicochemical condition of the system.

To address the time and cost limitations of ICP-OES for analyzing ionic mixtures, this work presents soft sensors that estimate ion concentrations in real time by leveraging easy-to-measure electrical conductivity data from conductivity sensors and mathematical models that map conductivity measurements to concentrations. Soft sensors are computer-based modeling frameworks that use data from hardware sensing devices to estimate process variables that are difficult to measure directly. The hardware sensor provides measurements of easily accessible variables that are mathematically related to the unmeasurable variable of interest. Soft sensors have been widely applied in chemical engineering for process monitoring, control, and fault detection, where robust mathematical models are essential for reliable state estimation. Systematic frameworks for soft sensor development have been proposed to enhance model robustness and applicability, incorporating structured steps for variable selection, model construction, and validation [4]. De Assis and Maciel Filho [5] used a soft sensor to estimate the unmeasurable states (e.g., biomass concentration) of a bioreactor that converts substrate into cell mass via mechanistic and empirical modeling (e.g., elemental balances). Furthermore, multivariate statistical techniques, including principal component analysis [6] and sensor validity index [7], have been employed for sensor fault detection in dynamic processes, improving process reliability and monitoring performance. These studies demonstrate the versatility of soft sensors across diverse engineering systems for state estimation, quality prediction, and fault diagnosis.

In this study, conductivity sensors are employed as hardware sensing devices within the soft sensor framework. Conductivity sensors measure the ionic conductivity of solutions by

applying an electrical current and measuring the resulting potential difference. Because ionic conductivity is governed by both the concentration and mobility of charge carriers (i.e., ions), fluctuations in ion concentration lead to corresponding changes in conductivity [8], enabling conductivity sensors to function as indirect measurement devices for ion concentration. Unlike ICP-OES, which requires offline sample handling, conductivity sensors are immersed directly in the solution, providing in situ conductivity measurements. Furthermore, due to the rapid nature of charge transport, they offer a promising pathway for real-time concentration monitoring via soft sensors. Many studies including Muetzel et al. [1], Ouimet et al. [9], Kamcev et al. [10], and Tweedie et al. [11] employed conductivity sensors to measure the dynamic evolution of ionic conductivity in membrane separation systems. Such measurements are critical for characterizing membrane feed and product streams and evaluating separation performance in real time.

Converting conductivity measurements into individual ion concentrations in complex multicomponent solutions remains a challenge. When a conductivity sensor applies an electric field, all ionic species contribute to the measured voltage used to compute the conductivity [12, 13]. As a result, the measured conductivity reflects the combined contribution of all dissolved ions, making it difficult to determine the concentration of individual species. Moreover, different ions possess distinct physical properties, such as diffusion coefficients and charge densities, which must be accurately accounted for when inferring concentrations from conductivity measurements. To overcome these challenges, mathematical models that relate ion concentrations to conductivity are embedded within the soft sensor framework to decouple the contributions of individual ions and enable rapid, low-cost concentration measurements.

Most studies [9, 14] use a polynomial model fitted to experimental data through linear regression or other data-driven approaches to predict conductivity measurements. For example, Smith et al. [14] fitted many polynomial models to the conductivity data of four ionic species present in the drainage of acid mines. The complexity of this modeling approach explodes as the number of species in the system increases. In addition, the mathematical structure of such models changes with ionic composition, with parameters that lack physical meaning. Although this approach may provide good conductivity predictions within the scope of the study, the models do not help to understand the physical system because the model parameters provide no insight on the behavior of ions under the applied electric field generated by the conductivity sensor. A quantitative understanding of ion transport in the presence of an electric field is essential to elucidate relative ion mobilities, inter-ionic

interactions, and their effects on electrical conductivity. Such physical insights also enable the optimization and control of key process parameters. Furthermore, purely data-driven models lack generalizability and cannot be readily extended to ionic systems containing new species or contaminants, as they typically perform poorly outside of the domain (or scope) of their training data [15, 16]. Consequently, changes in feed conditions (e.g., temperature or concentration) necessitate repeated model regression to capture new process dynamics, a procedure that is both tedious and time-consuming. For soft sensors involving real-time estimation, this modeling technique may be inefficient due to model selection and regression overhead. These limitations can be mitigated through the use of physics-based models that explicitly account for the key mechanisms governing ion transport. These physical models are at the heart of effective concentration soft sensors.

In the presence of an applied electric field, ion transport is primarily driven by the electrical force acting on the charged species. At finite concentrations, this transport is modified by second-order ion interaction effects, namely electrophoretic (or hydrodynamic) and relaxation effects, which are discussed in greater detail in Section 2. Under dilute conditions (less than 0.001 N), where these interaction effects are negligible, Debye-Hückel-Onsager (DHO) theory accurately describes the equivalent conductivity of single salt solutions, assuming complete dissociation of the dissolved salt into point-charged ions (i.e., without finite size) [17–21]. However, at concentrations exceeding 0.001 N, DHO theory no longer adequately captures experimental observations, as ionic interactions become increasingly significant [22].

Shedlovsky [22–24] proposed an equation that accounts for finite ion size and ion interactions, enabling accurate conductivity predictions for single salt solutions at concentrations up to 0.1 N. Nevertheless, for multi-salt solutions (i.e., solutions containing more than one dissolved salt), both the DHO and Shedlovsky theories become unreliable, as their governing equations are derived under the assumption that only two distinct and oppositely charged ions are present in solution. Naseri et al. [25, 26], Roger et al. [27], and Durand-Vidal et al. [28] derived the mean spherical approximation (MSA) model, which predicts the conductivity of solutions containing one or more dissolved salts (at ionic strengths up to 4 M) while accounting for complete or incomplete dissociation into charged, hard-sphere interacting ions.

The Shedlovsky and MSA models have been used in a few studies, including Robinson and Stokes [22], Naseri et al. [25, 26], and Roger et al. [27], to predict the conductivity of simple ionic solutions; however, their application to predicting ionic conductivity in membrane separation processes has not been reported. This is likely due to their mathematical

complexity and the lack of practical guidance regarding model selection, implementation, and limitations for membrane-relevant systems. To address this gap, to the best of our knowledge, this work presents the first systematic comparison of the Shedlovsky and MSA models to represent solution conductivity and to estimate ion concentrations in membrane feed/permeate solutions. Another novel aspect of this work is the extension of the Shedlovsky formulation from single salt systems to multi-salt mixtures. Therefore, two major questions are being addressed:

1. How accurate are the Shedlovsky and MSA models in predicting conductivity measurements in membrane-based separations?
2. Can the concentration of ions in multicomponent systems be accurately estimated using Shedlovsky and MSA-based soft sensors?

The Shedlovsky and MSA equations are implemented in the Python programming language to demonstrate how computational tools can be used to interrogate complex models. Their predictive performance is validated using conductivity data from single, binary, and ternary salt feed solutions containing sodium (Na^+), calcium (Ca^{2+}), and lanthanum (La^{3+}) ions, formulated to mimic the ionic composition of electronic waste leachates derived from cathode ray tube (CRT) and organic light-emitting diode (OLED) devices. For brevity, the term electronic waste hereafter refers specifically to CRT and OLED leachates. Throughout this study, all dissolved salts are assumed to fully dissociate into their respective ionic species. Model selection in multicomponent environments (e.g., ternary salt systems) is performed using the Hunter and Reiner [29] design criterion to effectively discriminate between the Shedlovsky and MSA models. Finally, ion concentration estimates generated by the soft sensor framework are validated using conductivity and pH measurements from the permeate outlet during the membrane separation of cobalt ions (Co^{2+}) from an acidic CoCl_2 feed representative of lithium-ion battery leachate.

This work is organized as follows: Section 2 presents the governing equations for conductivity measurements in ionic environments. Section 3 describes the experimental methods used to collect data for validating the soft sensor predictions of conductivity and ion concentration. Section 4 presents the performance results of the soft sensor to predict the conductivity of single, binary, and ternary salt solutions and to estimate real-time ion concentrations from conductivity data obtained during membrane processes. Finally, Section 5 discusses the conclusions and future opportunities.

2. Modeling and Theory

2.1. Conductivity sensors and the governing equations for conductivity measurements

The conductivity sensor used in this study employs a six-channel configuration for conductivity measurements (Figure 1). A controlled electrical current passes through the ionic solution, and the potential difference created by the current flow is measured to determine the conductivity of the solution, as defined in Eq. (3).

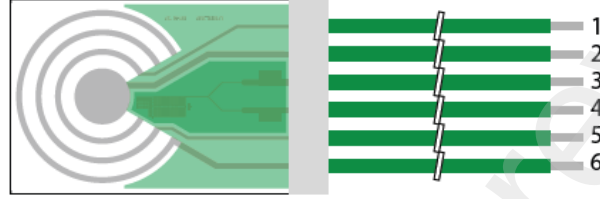


Figure 1: Innovative Sensor Technology LFS 1107 Conductivity Sensor [30]. Channels 2 and 6 allow the passage of electrical current through the solution. Channels 1 and 5 measure the potential difference created by the current flow, while channels 3 and 4 enable temperature measurements and compensation.

Eq. (3) is derived from the relationship between conductivity and resistivity, with conductivity (χ) defined as the inverse of resistivity (ρ):

$$\chi = \frac{1}{\rho} \quad (1)$$

and resistivity is defined as:

$$\rho = \frac{RA}{l} \quad (2)$$

Substituting Eq. (2) and $R = \frac{V}{I}$ into Eq. (1) yields,

$$\chi = \frac{l}{RA} = \underbrace{\frac{l}{A}}_{\Phi} \frac{I}{V} = \Phi \frac{I}{V} \quad (3)$$

where χ is the conductivity in $\text{S} \cdot \text{cm}^{-1}$, l is the effective distance between the voltage-sensing channels in cm, A is the effective cross-sectional area through which the current flows in cm^2 , R is the resistance in Ω (ohm), I is the applied current in A (ampere), and V is the measured voltage in V (volt).

The term Φ is a constant (commonly referred to as cell constant) in cm^{-1} because the geometry (e.g., the area) of the conductivity sensor is constant.

There are two types of conductivity measurement: (1) specific conductivity χ , usually referred to as conductivity, and (2) equivalent conductivity (Λ), which represents the conductivity per equivalent of ion or salt. Miller et al. [8] established a relationship between χ and Λ :

$$\Lambda = \frac{1000\chi}{c_{\text{eq}}}, \quad \text{with} \quad c_{\text{eq}} = z_i c_i \quad (4)$$

where Λ is the equivalent conductivity in $\text{cm}^2 \cdot \text{S} \cdot \text{equiv}^{-1}$, c_{eq} is the salt concentration in N ($\text{equiv} \cdot \text{L}^{-1}$), c_i is the concentration of ionic species i in M ($\text{mol} \cdot \text{L}^{-1}$), and z_i is the valency of ionic species i .

Therefore, the conductivity of a salt or ionic solution can be calculated from the concentration as:

$$\chi = \frac{\Lambda c_{\text{eq}}}{1000} = \frac{z_i \Lambda c_i}{1000} \quad (5)$$

2.2. The Shedlovsky model for predicting single salt conductivity from concentration

At very low concentrations (less than 0.001 N), the equivalent conductivity of fully dissociated salt solutions decreases linearly with the square root of the concentration [22, 31]:

$$\Lambda = \Lambda^\circ - K\sqrt{c} \quad (6)$$

where Λ° is the limiting equivalent conductivity of the dissolved salt in $\text{cm}^2 \cdot \text{S} \cdot \text{equiv}^{-1}$, K is an empirical constant that is fitted to the data, and c is the salt concentration in M.

The limiting equivalent conductivity Λ° , is the characteristic conductivity of a salt at infinite dilution, a state in which the ions are far apart so that their interactions with each other are negligible [22, 32, 33].

Onsager and Fuoss [34], Debye and Hückel [17], Onsager [18], and Fuoss and Onsager [19] developed a theoretical framework (DHO theory) describing the relationship observed in Eq.(6) based on ion interaction theory. This framework assumes that dissolved salts fully dissociate into ions treated as point charges with no finite size, each surrounded by a spherical symmetric cloud of opposite charge.

Under the influence of the electric field generated by the conductivity sensor, these ions become mobile, diffusing and interacting with each other, giving rise to two dominant effects: electrophoretic (or hydrodynamic) and relaxation effects [22, 35, 36]. The motion of an ion induces solvent flow, which in turn affects the mobility of neighboring ions. Consequently, ions move through a disturbed medium, an effect known as the electrophoretic effect [22, 37]. The mobility of ions induced by the external electric field distorts the symmetrical

distribution of ions at equilibrium [22]. At equilibrium, there is no net electric force acting on the central ion; however, when an external electric field is applied, the central ion may begin to drift in the direction of the field. Instantly, the central ion experiences an electrostatic pull from ion clouds with opposite charges, a phenomenon known as the relaxation effect [22, 38].

The DHO theory is presented as follows:

$$\Lambda = \Lambda^\circ - (B_1 \Lambda^\circ + B_2) \sqrt{c} \quad (7a)$$

with

$$B_1 = \frac{2.801 \times 10^6 |z_1 z_2| q}{(\varepsilon_r T)^{3/2} (1 + \sqrt{q})}, \quad (7b)$$

$$B_2 = \frac{41.25(|z_1| + |z_2|)}{\eta(\varepsilon_r T)^{1/2}}, \quad (7c)$$

and

$$q = \frac{|z_1 z_2|}{|z_1| + |z_2|} \cdot \frac{\lambda_1^\circ + \lambda_2^\circ}{|z_2| \lambda_1^\circ + |z_1| \lambda_2^\circ}. \quad (7d)$$

where λ_i° is the limiting equivalent conductivity of ion i in $\text{cm}^2 \cdot \text{S} \cdot \text{equiv}^{-1}$, B_1 and B_2 are coefficients of the relaxation and electrophoretic terms, respectively, z_i is the valency of ion i , c is the salt concentration in M, T is the temperature in K, ε_r is the relative permittivity of the solvent, and η is the viscosity of the solvent in P (Poise).

However, at concentrations above 0.001 N, Eq. (7a) fails to accurately predict equivalent conductivity [22], because at these concentration levels, the assumption of infinite dilution and no finite ion size breaks down as ions become more closely packed.

Shedlovsky [22–24, 39] extended Eq. (7a) to account for the finite size of ions to more accurately represent equivalent conductivity data at concentrations up to 0.1 N:

$$\Lambda = \Lambda^\circ - \frac{\sqrt{I}}{1 + 50.29 \times 10^8 (\varepsilon_r T)^{-0.5} a \sqrt{I}} A, \quad (8a)$$

with

$$I = \frac{1}{2} \sum_{i=1}^s c_i z_i^2, \quad (8b)$$

$$A = B_1 \Lambda^\circ + B_2. \quad (8c)$$

where s is the number of ions in the solution, I is the ionic strength of the solution in M, c_i

is the concentration of ion i in M, and a is the distance from an ion within which no other ion can penetrate in cm. All other parameters remain the same as defined in Eq. (7a).

Eq. (8a) can be substituted into Eq. (4), leading to Eq. (9):

$$\chi = \frac{c_{eq}}{1000} \left(\Lambda^\circ - \frac{\sqrt{I}}{1 + Ba\sqrt{I}} A \right) = \frac{z_i c_i}{1000} \left(\Lambda^\circ - \frac{\sqrt{I}}{1 + Ba\sqrt{I}} A \right) \quad (9)$$

where the lumped parameter $B = 50.29 \times 10^8 (\epsilon_r T)^{-0.5}$.

2.3. The extended Shedlovsky model for predicting multi-salt conductivity from concentration

The Shedlovsky formulation presented in Section 2.2 has traditionally been restricted to single salt systems, and its application to multi-salt environments has received limited attention. However, most feed solutions encountered in separation processes contain multiple dissolved salts. In this work, we extend the applicability of the Shedlovsky model to multi-salt solutions by treating each salt contribution independently and superimposing their respective conductivity contributions within a unified framework. In implementing this extension, and following approaches similar to those of Naseri et al. [25, 26], Roger et al. [27], and Durand-Vidal et al. [28, 40], we assume that the closest distance of approach parameter, a , for the multi-salt system can be approximated by the mean value of the corresponding parameters for the individual dissolved salts. This assumption provides a practical means of retaining a single effective a parameter while preserving the theoretical structure of the original single salt formulation.

The proposed extension allows the Shedlovsky model to be incorporated into soft sensor architectures for realistic process streams, thereby bridging the gap between classical electrolyte theory and practical membrane separation applications. Under this extended framework, the overall conductivity of a multi-salt system can be expressed as follows:

$$\chi = \sum_{j=\text{salt } 1}^{\text{salt } m} \chi_j \quad (10a)$$

$$\chi = \sum_{j=\text{salt } 1}^{\text{salt } m} \frac{c_{eq,j}}{1000} \left(\Lambda_j^\circ - \frac{\sqrt{I_j}}{1 + Ba_{avg}\sqrt{I_j}} A_j \right) \quad (10b)$$

with

$$a_{avg} = \frac{1}{m} \sum_{j=\text{salt } 1}^{\text{salt } m} a_j \quad (10c)$$

where m is the number of salts in the solution.

We acknowledge that the assumption of non-interacting dissolved salts introduces limitations to the extended formulation. As the number of salts in solution increases, the neglect of cross-ionic interactions, particularly repulsive cation–cation interactions, may lead to increased prediction error. Consequently, for systems beyond single salt solutions, we expect the extended Shedlovsky model to progressively overpredict conductivity due to the omission of these interaction effects.

Section 2.4 describes a more comprehensive model that explicitly accounts for all ionic interactions in multi-salt systems. The extended Shedlovsky formulation presented in Eq. (10a) is therefore intended to serve as a simplified predictive framework for multi-salt conductivity and as a foundational step toward generalizing the classical Shedlovsky approach to more complex ionic environments. Throughout this work, the term Shedlovsky model is used to refer collectively to both the original single salt formulation and the extended multi-salt formulation, unless otherwise specified.

2.4. The mean spherical approximation model for predicting the conductivity of multi-salt systems

The mean spherical approximation (MSA) model as presented by Naseri et al. [25, 26], Roger et al. [27], and Durand-Vidal et al. [28, 40] calculates the electrical conductivity of multi-salt solutions as:

$$\chi = \frac{e^2}{k_B T} \sum_{i=1}^s n_i D_i^\circ z_i^2 \left(1 + \frac{\delta v_i}{v_i^\circ}\right) \left(1 + \frac{\delta k_i}{k_i}\right) \quad (11)$$

where s is the number of ions in the solution, n_i is the number density of species i , e is the elementary charge, k_B is the Boltzmann constant, T is the temperature of the solution, D_i° is the diffusion coefficient of species i at infinite dilution, $\delta v_i/v_i^\circ$ is the electrophoretic correction on the velocity of species i , and $\delta k_i/k_i$ is the relaxation correction on the electric force exerted on species i . All parameters are in the International System of Units.

Naseri et al. [25, 26] developed equations for electrophoretic and relaxation corrections in Eq. (11). The electrophoretic correction is given by:

$$\frac{\delta v_i}{v_i^\circ} = \frac{F^2 |z_i|}{12\pi\epsilon_o\epsilon_r\eta k_B T N_A \lambda_{\text{mol},i}^\circ} \sum_{j=1}^s n_j e^2 z_j^2 \left[\frac{\exp(\kappa(\sigma_i - \sigma_{ij}))}{\kappa(1 + \kappa\sigma_i)} + \frac{\exp(\kappa(\sigma_j - \sigma_{ij}))}{\kappa(1 + \kappa\sigma_j)} \right] \quad (12a)$$

with

$$\kappa = \sqrt{\frac{e^2}{\varepsilon_o \varepsilon_r k_B T} \sum_{j=1}^s n_j z_j^2} \quad (12b)$$

and

$$\sigma_{ij} = \frac{\sigma_i + \sigma_j}{2} \quad (12c)$$

where κ is the Debye length, F is the Faraday's constant, ε_o is the permittivity of free space, ε_r is the relative permittivity of the solvent, η is the viscosity of the solvent, N_A is the Avogadro's number, σ_i is the hard sphere diameter of species i , and $\lambda_{\text{mol},i}^\circ$ is the limiting molar conductivity of species i . Again, the parameters are in the International System of Units.

The Debye length κ characterizes the distance over which the electrostatic influence of an ion is screened by the surrounding medium [41, 42]. The relaxation correction is calculated as:

$$\begin{aligned} \frac{\delta k_i}{k_i} = & \frac{e z_i}{24 \pi \varepsilon_o \varepsilon_r k_B T} \sum_{p=1}^s \left[\frac{\zeta_i^p}{\sqrt{q_p} + q_p} \sum_{j=1}^s \sum_{r=1}^s \left(\frac{t_j \zeta_j^p \mu_r (e_r \omega_r - e_j \omega_j) \sinh(\kappa \sqrt{q_p} \sigma_{jr})}{\omega_r - \omega_j} \frac{1}{\sigma_{jr}} \right) \right. \\ & \times \left. \left[\frac{\exp(\kappa \sigma_j)}{1 + \kappa \sigma_j} + \frac{\exp(\kappa \sigma_r)}{1 + \kappa \sigma_r} \right] \exp \left[-\kappa (1 + \sqrt{q_p}) \sigma_{jr} \right] \right] \end{aligned} \quad (13a)$$

with

$$t_j = \frac{\mu_j \omega_j}{\bar{\omega}}, \quad (13b)$$

$$\omega_r = \frac{D_r^\circ}{k_B T} = \frac{N_A \lambda_{\text{mol},r}^\circ}{F^2 |z_r|}, \quad (13c)$$

$$\mu_r = \frac{n_r e_r^2}{\sum_{m=1}^s n_m e_m^2}, \quad (13d)$$

$$\bar{\omega} = \sum_{m=1}^s \mu_m \omega_m, \quad \text{and} \quad e_r = e z_r \quad (13e)$$

where t_j is the transport number of species j , μ_r is the relative ionic strength of species r , ω_r is the ionic mobility of species r , and $\bar{\omega}$ is the average mobility.

The transport number t and ionic mobility ω measure the fraction of the total electric current carried by ions and the speed of migration of ions under the influence of the external

electric field, respectively. q and ζ in Eq. (13a) are defined by Eq. (14a) and Eq. (15a), respectively:

$$q_p = \sum_{m=1}^s \frac{\bar{\omega} t_m}{\omega_m + \alpha_p} \quad (14a)$$

with

$$-2\bar{\omega}\alpha \sum_{m=1}^s \frac{t_m}{\omega_m^2 - \alpha^2} = 0 \quad (14b)$$

and

$$\zeta_j^p = \frac{N_p \omega_j}{\omega_j^2 - \alpha_p^2} \quad (15a)$$

with

$$\frac{1}{N_p^2} = \sum_{m=1}^s \frac{t_m \omega_m^2}{(\omega_m^2 - \alpha_p^2)^2} \quad (15b)$$

Onsager [43] established that the roots of Eq. (14b) are bounded by the mobility of the ions.

$$0 = \alpha_1^2 < \omega_1^2 < \alpha_2^2 < \omega_2^2 < \dots < \alpha_s^2 < \omega_s^2 \quad (16)$$

This property is employed in subsequent work by Naseri et al. [26], Roger et al. [27], Durand-Vidal et al. [28], and Van Damme and Deconinck [44].

Table 1 and 2 report the values for the Shedlovsky and MSA model parameters used in the case studies to predict conductivity measurements.

2.5. Computational implementation of the Shedlovsky and MSA conductivity models

The Shedlovsky and MSA conductivity models were implemented in `Python` version 3.10.14 on a Lenovo Yoga 7i, Windows 11 Home operating system, with an Intel Core i7-1355U (1.70 GHz) processor, and 16.0 GB of RAM.

The root of Eq. (14b) was determined by interval bisection using the function `bisect` in `Scipy`. The hyperbolic sine, square root, absolute value, and exponential functions were calculated using `math` and `numpy` functions.

The Shedlovsky and MSA models are available at <https://github.com/dowlinglab/salts-conductivity-modelling>

3. Experimental Methodology

This section presents the methods and equipment used to collect conductivity and concentration data to validate the soft sensors.

Species	Λ° (cm ² · S · equiv ⁻¹)	Species	$a \times 10^8$ (cm)
NaCl	126.45	NaCl	4.0
KCl	149.85	KCl	4.6
HCl	426.15	HCl	4.4
LiCl	114.95	LiCl	5.2
CaCl ₂	135.85	CaCl ₂	4.3
CoCl ₂	131.35	CoCl ₂	4.5
LaCl ₃	145.90	LaCl ₃	4.9
Na ⁺	50.10		
K ⁺	73.50		
H ⁺	349.80		
Li ⁺	38.60		
Ca ²⁺	59.50		
Co ²⁺	55.00		
La ³⁺	69.70		
Cl ⁻	76.35		

Table 1: Limiting equivalent conductivity (Λ°) at 25°C and distance of closest approach (a) of ionic species studied in this work [22, 39].

Species	$\lambda_{\text{mol}}^\circ \times 10^4$ (S · m ² · mol ⁻¹)	$\sigma \times 10^{10}$ (m)	$D^\circ \times 10^9$ (m ² · s ⁻¹)
Na ⁺	50.08	2.04	1.33
K ⁺	73.48	2.76	1.96
H ⁺	349.70	0.74	9.31
Li ⁺	38.66	1.48	1.03
Ca ²⁺	59.47	2.00	0.79
Co ²⁺	55.00	1.50	0.73
La ³⁺	69.70	2.36	0.62
Cl ⁻	76.31	3.62	2.03

Table 2: Limiting molar conductivity ($\lambda_{\text{mol}}^\circ$) and diffusion coefficient at infinite dilution (D°) at 25°C and hard sphere diameter (σ) of ionic species studied in this work [25, 45].

3.1. Conductivity sensor measurements of feed solutions to evaluate the predictive performance of the Shedlovsky and MSA models

The first validation case considers feed solutions containing chloride salts commonly found in electronic waste, such as sodium chloride (NaCl), calcium chloride (CaCl₂), and lanthanum chloride (LaCl₃). Experimental data include conductivity measurements for single, binary, and ternary combinations of these species at 25°C over a range of concentrations (or ionic strengths), as shown in Figure 3, 4, and 6. A single salt potassium chloride (KCl) solution is used as a reference to validate the measurements of the conductivity sensor against standard values from the National Institute of Standards and Technology (NIST).

Three primary stock solutions—100 mM NaCl, 100 mM CaCl₂, and 50 mM LaCl₃—were prepared by dissolving 2.92 g, 5.55 g, and 6.13 g of the respective solid state salts in 500 mL of deionized (DI) water sourced from a Millipore water purification system (Milli Q Advantage A10). A 10 mL sample of single salt solutions was obtained at specific concentrations by diluting the required volume of the stocks in the corresponding volume of DI water using the standard dilution equation ($c_1V_1 = c_2V_2$). A similar procedure was used to prepare 10 mL samples of binary salt mixtures at various molar ratios (e.g., 1:1, 1:2, 2:1, and 3:1). Precise volume transfers were performed using 1 and 5 mL Eppendorf pipettes ($\pm 0.6\%$).

Electrical conductivity was measured using an Innovative Sensor Technology LFS 1107 Conductivity Sensor (Figure 1) interfaced with an EC Evaluation Board LFS. To ensure measurement stability, the conductivity sensor was submerged in each sample for 10 seconds prior to data acquisition. The conductivity readings were recorded at room temperature. Between measurements, the sensor was thoroughly rinsed with DI water. To ensure reproducibility, experiments were conducted in duplicate using two independent conductivity sensors, each with a unique cell constant (as defined in Eq. (3) and listed in Table S1), designated as Probe 1 and Probe 2. The data acquired from Probe 1 are presented in Section 4, with additional uncalibrated results for both Probe 1 and Probe 2 provided in Section S1.

3.2. Diafiltration membrane data to validate real-time ion concentration estimates from the soft sensors

A diafiltration apparatus (Figure 2) for the separation of high-value cations (e.g., Co²⁺) from aqueous solutions (e.g., acidic CoCl₂ representative of lithium-ion battery leachate) is used to validate the soft sensor estimates of ion concentrations in the feed/permeate. Conductivity sensors are installed on the feed and permeate sides of the apparatus to provide real-time conductivity measurements for the soft sensor framework [9, 46].

The diafiltration experiments were conducted using a custom-built apparatus (Figure 2) consisting of a diafiltrate reservoir connected to a 3D-printed stirred cell (10 mL volume). The system was operated under concentrating-lag diafiltration conditions and maintained at a constant transmembrane pressure of 60 psi (4.14 bar) throughout the experiment [9].

Inline conductivity sensors (LFS 1107 Conductivity Sensor from Innovative Sensor Technology, Las Vegas, Nevada) were installed at both the retentate and permeate outlets to enable continuous, real-time monitoring of solution conductivity. System pressure was monitored using a pressure transducer (Omega PX409-USBH). Permeate mass was recorded using an analytical balance mounted with an automated rotating vial collection system, allowing synchronized measurement of permeate mass during the experiment [46, 47].

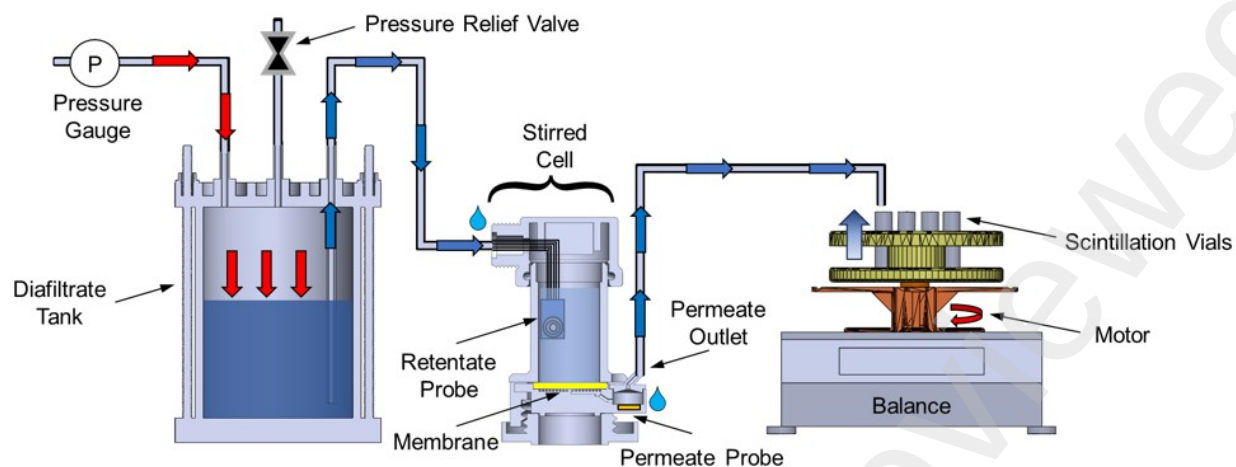


Figure 2: Diafiltration apparatus for the separation of target cations from aqueous solutions. The retentate and permeate probes track the conductivity of the retentate and permeate outlets, respectively [1, 9, 46].

Single salt diafiltration experiments were performed using cobalt chloride (Sigma-Aldrich, St. Louis, MO, USA). A commercial polyamide nanofiltration membrane (Veolia DK) was used for the experiment. Circular membrane coupons (1 inch diameter) were soaked in deionized water (Milli Q Advantage A10) overnight prior to use and subsequently compacted with deionized water at 60 psi for 3 hours to achieve stable water flux.

The initial feed solution contained 1 mM CoCl_2 , while the diafiltrate reservoir contained 50 mM CoCl_2 . Both solutions were adjusted to pH 3 (1 mM H^+) using hydrochloric acid (HCl). During diafiltration, the system was operated such that one drop of diafiltrate was added to the stirred cell for every drop of permeate removed, maintaining a constant liquid volume in the stirred cell. Permeate was collected sequentially in ten aliquots of 1.5 g each, enabling time-resolved evaluation of Co^{2+} transport under single salt diafiltration conditions.

Continuous conductivity data from the inline sensors were recorded throughout the experiment. In addition, discrete permeate samples were collected for chemical analysis (Figure 8 and 9). Permeate pH was measured immediately after the completion of each experiment using a portable pH meter (Thermo Scientific Accumet AP115) to minimize the effects of atmospheric CO_2 dissolution. Co^{2+} concentrations in feed, diafiltrate, and permeate samples were quantified using ICP-OES.

3.3. Experimental data error analysis

Data precision and reliability were quantified by collecting duplicate conductivity measurements from a single salt KCl solution prepared according to the experimental procedure described in Section 3.1 using Probe 1 and Probe 2. The average conductivity (χ_{avg}) of the

measurements obtained from both probes was calculated for each experimental condition (i.e., salt concentration) as

$$\chi_{\text{avg},r} = \frac{\chi_{\text{Probe } 1,r} + \chi_{\text{Probe } 2,r}}{2}, \quad \forall r \in \{1, 2, \dots, N_{\text{exp}}\} \quad (17)$$

where N_{exp} is the number of experimental conditions.

For each probe, the measurement error was quantified as the mean absolute percent error (MAPE) between the individual probe readings and the corresponding average conductivity value across all experimental conditions:

$$\sigma_{\chi, \text{Probe } i} = \frac{1}{N_{\text{exp}}} \sum_{r=1}^{N_{\text{exp}}} 100 \frac{|\chi_{\text{avg},r} - \chi_{\text{Probe } i,r}|}{\chi_{\text{avg},r}}, \quad \forall i \in \{1, 2\} \quad (18)$$

where σ_{χ} represents the measurement error of the probes expressed as a percentage of the conductivity measurement.

The calculated error values were then used to generate error bands for the conductivity data. The upper (χ^+) and lower (χ^-) bounds for each probe measurement were computed as

$$\chi_{\text{Probe } i}^+ = \chi_{\text{Probe } i} + \sigma_{\chi, \text{Probe } i} \chi_{\text{Probe } i}, \quad \chi_{\text{Probe } i}^- = \chi_{\text{Probe } i} - \sigma_{\chi, \text{Probe } i} \chi_{\text{Probe } i}, \quad \forall i \in \{1, 2\} \quad (19)$$

Using conductivity measurements of the single salt KCl solution (Section S1), the measurement error for both probes was determined to be 0.93% of the measured conductivity values. Similarly, by defining the MAPE for current and voltage readings, the measurement errors associated with the applied electrical current and the measured voltage of both probes were calculated to be 1.00% and 0.14%, respectively.

4. Results and Discussion

This section presents the results from the two soft sensors (Shedlovsky-based and MSA-based) and the associated case studies.

4.1. Shedlovsky and MSA models accurately predict the conductivity of single salt solutions containing ions typical of electronic waste

Figure 3 shows the conductivity predictions of the Shedlovsky and MSA models in feed solutions containing ions commonly found in electronic waste. Table 3 quantifies the predictive accuracy of both models using MAPE, mean squared error (MSE), and mean absolute

error (MAE) metrics. As shown in Figure 3 and Table 3, using the parameters listed in Table 1 and 2, both Shedlovsky and MSA models accurately predict conductivity measurements of single salt KCl, NaCl, CaCl₂, and LaCl₃ solutions prepared according to the experimental procedures described in Section 3.1, with MAPE values below 2.3%. Because both models capture all the ionic interactions in single salt systems, they yield a similar prediction accuracy, as expected.

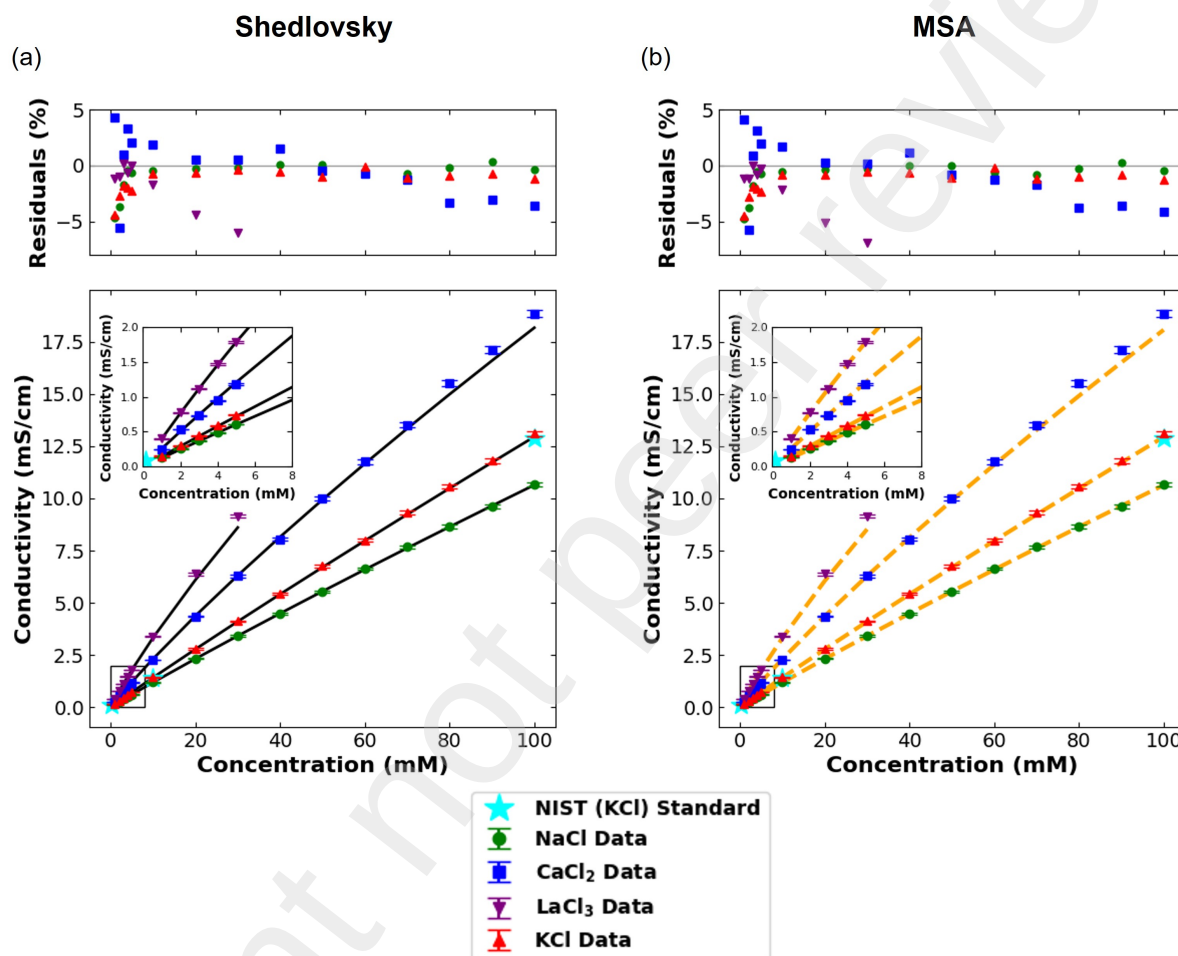


Figure 3: Shedlovsky (a) and MSA (b) models prediction for the conductivity of single salt feed solutions typical of electronic waste. The top panel of each subplot presents the percent errors of the model predictions. The inset (top-left) in the bottom panel of each subplot highlights data in the low concentration region. The data (KCl) obtained from the conductivity sensor is consistent with National Institute of Standards and Technology (NIST) reference values.

The observed increase in prediction errors when transitioning from Na⁺ to Ca²⁺ and La³⁺ systems is attributed to the intensified electrostatic interactions associated with higher cationic charge density. As electrostatic forces scale with the square of the ionic charge,

according to Eq. (12b), divalent and trivalent ions exhibit significantly stronger interaction effects than monovalent species. Consequently, any inaccuracies in the representation of mobility corrections, electrostatic interactions, or effective ion size are amplified with increasing cation valence. Both the Shedlovsky and MSA models treat ions as finite-sized species with fixed diameters (or closest-approach distances), an assumption that may be less appropriate for strongly hydrated multivalent ions, whose effective size and solvation structure can vary with concentration [22, 48]. In addition, solvent effects such as dielectric decrement, which are often treated approximately (or neglected) in practical implementations of these models, may become increasingly important at higher ionic strengths [49, 50]. These combined factors likely contribute to the larger underprediction observed for the CaCl_2 and particularly the LaCl_3 systems.

Model	MSE ($\mu\text{S} \cdot \text{cm}^{-1}$) ²	MAE ($\mu\text{S} \cdot \text{cm}^{-1}$)	MAPE (%)
NaCl			
Shedlovsky	512	16	1.0
MSA	614	17	1.0
CaCl_2			
Shedlovsky	70,215	156	2.2
MSA	94,670	179	2.3
LaCl_3			
Shedlovsky	48,027	114	1.9
MSA	63,465	133	2.2
KCl			
Shedlovsky	3,820	43	1.4
MSA	4,475	47	1.5

Table 3: Error metrics for the conductivity predictions in $\mu\text{S} \cdot \text{cm}^{-1}$ of the Shedlovsky and MSA models in single salt feed solutions typical of electronic waste, where the lowest values are bold.

The predictive performance of the Shedlovsky and MSA models was also evaluated for acidic aqueous solutions, specifically lithium-ion battery leachate; other process streams (e.g., mine drainage, coal refuse) can also exhibit similar acidic conditions [51–54]. The ionic species considered in this study include Li^+ , Co^{2+} , and H^+ . Figure S2 shows that both models accurately predict the conductivity of single salt LiCl , CoCl_2 , and HCl solutions (prepared using a procedure similar to that described in Section 3.1), with MAPE values below 6%, as presented in Table S2.

Based on the results of both case studies, the Shedlovsky and MSA models provide accurate conductivity predictions for single salt systems across a range of monovalent, divalent, and trivalent cations. The comparable predictive performance indicates that either model may be employed for the estimation of single salt conductivity within the concentration ranges examined in this study. However, when computational efficiency is a primary consideration, the Shedlovsky model offers a practical advantage due to its simpler mathematical structure. Therefore, for single salt systems, where both models achieve similar accuracy, the Shedlovsky model represents the more efficient choice for real-time process monitoring applications.

4.2. Shedlovsky and MSA models validate the conductivity of binary salt solutions representative of electronic waste

Figure 4 presents the conductivity predictions of the Shedlovsky and MSA models for binary salt systems ($\text{NaCl} - \text{CaCl}_2$, $\text{NaCl} - \text{LaCl}_3$, and $\text{CaCl}_2 - \text{LaCl}_3$) representative of electronic waste. The feed solutions were prepared according to the experimental procedures described in Section 3.1 at various concentration ratios (e.g., 1:1, 1:2).

Using the parameters listed in Table 1 and 2, both models show good agreement with the conductivity measurements. Repeating MSE, MAE, and MAPE error metrics, Table 4 shows that both models accurately predict the conductivity of binary salt feed solutions with MAPE values below 5%.

As shown in Table 4, the MAPE values associated with the Shedlovsky model increase as the total cation charge of the binary system increases. This trend indicates that prediction accuracy deteriorates in mixtures containing higher-valence ions.

The inferior performance of the Shedlovsky model relative to the MSA model for binary salt solutions can be attributed to its simplified treatment of dissolved salts. In the extended formulation, each salt contribution is evaluated independently and subsequently superimposed, such that cross-interactions are not explicitly represented. In particular, cation–cation correlation effects are neglected. As discussed in Section 2.3, these interactions may become increasingly significant as the overall cation charge density increases, and therefore their omission may lead to larger deviations from experimental conductivity measurements. The consistent positive percent residuals observed in Figure 4 indicate a systematic overprediction by the extended Shedlovsky model, consistent with the expected behavior described in Section 2.3.

Although the MSA model accounts for the cation-cation interactions not considered in the Shedlovsky model, it underpredicts the conductivity data (see Figure 4). This devia-

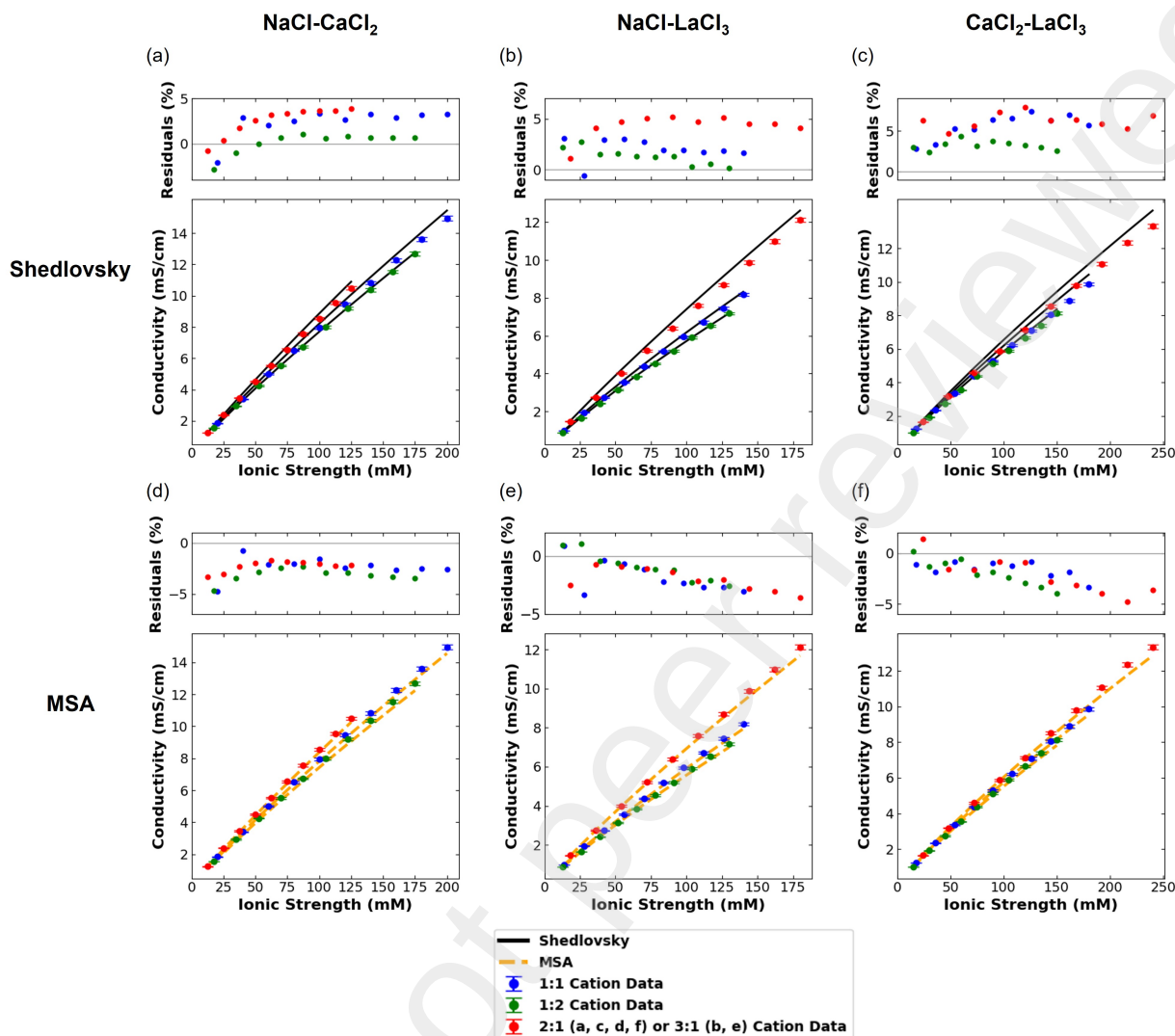


Figure 4: Shedlovsky and MSA models conductivity prediction for (a, d) NaCl – CaCl₂, (b, e) NaCl – LaCl₃, and (c, f) CaCl₂ – LaCl₃ binary salt solutions representative of electronic waste. The top panel of each subplot presents the percent errors of the model predictions.

tion likely arises from the simplifying assumptions, including fixed ionic diameters and a constant bulk dielectric constant (relative permittivity). In reality, the intense electric fields surrounding multivalent Ca²⁺ and La³⁺ ions can induce localized polarization of neighboring water molecules [22, 55]. This electrostriction effect may alter the local solvent structure and hydration shell, thereby enhancing ionic mobility and resulting in solution conductivities that exceed MSA predictions [55, 56].

The lower MAPE value of the Shedlovsky model in the NaCl – CaCl₂ system (see Table 4) is primarily driven by the 1:2 concentration ratio. At this ratio, the ionic strength contri-

Model	MSE ($\mu\text{S} \cdot \text{cm}^{-1}$) ²	MAE ($\mu\text{S} \cdot \text{cm}^{-1}$)	MAPE (%)
NaCl-CaCl ₂			
Shedlovsky	47,828	166	2.1
MSA	46,468	184	2.6
NaCl-LaCl ₃			
Shedlovsky	45,285	150	2.6
MSA	23,534	111	1.8
CaCl ₂ -LaCl ₃			
Shedlovsky	161,045	329	5.0
MSA	45,892	151	2.0

Table 4: Error metrics for the conductivity predictions in $\mu\text{S} \cdot \text{cm}^{-1}$ of the Shedlovsky and MSA models in binary salt solutions representative of electronic waste, where the lowest values are bold.

bution of Ca^{2+} is significantly higher than that of Na^+ , causing ion transport in the binary salt solution to behave like a single salt as Ca^{2+} and Cl^- dominate the transport numbers ($\text{Ca}^{2+} : 0.36$, $\text{Na}^+ : 0.07$, and $\text{Cl}^- : 0.57$). This behavior is less pronounced in the 1:1 and 2:1 ratios, where the ionic strength contributions of Ca^{2+} and Na^+ are more comparable.

Figure 4 and Table 4 show that the MSA model has higher predictive accuracy and consistency relative to the Shedlovsky model for the binary salt systems examined. Accordingly, when prediction accuracy is the primary performance criterion in binary salt environments, the MSA model represents the preferred framework. However, when computational efficiency is a dominant consideration, the Shedlovsky model offers a practical advantage due to its reduced mathematical complexity. The choice between models should therefore be guided by the specific accuracy and computational requirements of the intended application.

4.3. Shedlovsky and MSA models guide the collection of conductivity data for predicting multicomponent solutions

Membranes for the advanced separation of high-value metal ions from multicomponent feed solutions, including those derived from electronic waste, spent batteries, and other domestic sources, are being developed to support this study and related applications. The binary salt systems analyzed in Section 4.2 represent simplified cases involving the separation of two cations (e.g., Na^+ and Ca^{2+}) with a common anion (e.g., Cl^-). However, electronic and battery waste streams typically contain a larger number of dissolved cations, including species present as contaminants or impurities. Thus, the Shedlovsky and MSA models need

to be validated for more complex systems.

As such, conductivity data for ternary salt feed solutions (and beyond) are required to enable the development of modeling capabilities for real-time ion concentration measurements in complex ionic environments. Based on the binary salt conductivity results in Section 4.2, which demonstrate a higher prediction error for the Shedlovsky model relative to the MSA model due to neglected cation–cation interactions, a greater deviation from experimental data is anticipated for the Shedlovsky model in ternary salt systems. Therefore, at a given feed ionic strength, identifying salt concentrations that maximize the difference between the conductivity prediction of the Shedlovsky and MSA models is essential for effective model discrimination, as it enables the selection of the most appropriate model for multicomponent systems.

To this end, Figure 5 considers four ionic strengths of the ternary salt solution for model discrimination; 50, 100, 150, and 200 mM. For each of these ionic strengths, as defined in Eq. (20), the corresponding concentration of Ca^{2+} was calculated from the concentrations of Na^+ and La^{3+} , as defined in Eq. (22) and shown in Figure S3.

$$I = \frac{1}{2} (c_{\text{Na}^+} z_{\text{Na}^+}^2 + c_{\text{Ca}^{2+}} z_{\text{Ca}^{2+}}^2 + c_{\text{La}^{3+}} z_{\text{La}^{3+}}^2 + c_{\text{Cl}^-} z_{\text{Cl}^-}^2) \quad (20)$$

Assuming complete dissociation of the dissolved salts into their respective ionic species:

$$c_{\text{Cl}^-} = z_{\text{Na}^+} c_{\text{Na}^+} + z_{\text{Ca}^{2+}} c_{\text{Ca}^{2+}} + z_{\text{La}^{3+}} c_{\text{La}^{3+}} \quad (21)$$

Substituting Eq. (21) into Eq. (20) and solving for $c_{\text{Ca}^{2+}}$ yields:

$$c_{\text{Ca}^{2+}} = \frac{2I - c_{\text{Na}^+} z_{\text{Na}^+} (z_{\text{Na}^+} + z_{\text{Cl}^-}^2) - c_{\text{La}^{3+}} z_{\text{La}^{3+}} (z_{\text{La}^{3+}} + z_{\text{Cl}^-}^2)}{z_{\text{Ca}^{2+}} (z_{\text{Ca}^{2+}} + z_{\text{Cl}^-}^2)} \quad (22)$$

where I is the ionic strength of the ternary salt solution in mM, c_k is the concentration of the ionic species k in mM, and z_k is the valency of the ionic species k .

Once the concentrations of all the ionic species have been calculated from the desired ionic strength, the conductivity predictions of the Shedlovsky and MSA models were evaluated using Eq. (10b) and Eq. (11), respectively. The difference in the predictions of the Shedlovsky and MSA models is shown in Figure 5 and calculated as follows:

$$\text{Model Prediction Difference} = \frac{\chi_{\text{Shedlovsky}} - \chi_{\text{MSA}}}{\chi_{\text{MSA}}} \times 100\% \quad (23)$$

To effectively discriminate between the two models, it is necessary to identify the experimental conditions where their predictions diverge maximally, in accordance with the design criterion proposed by Hunter and Reiner [29].

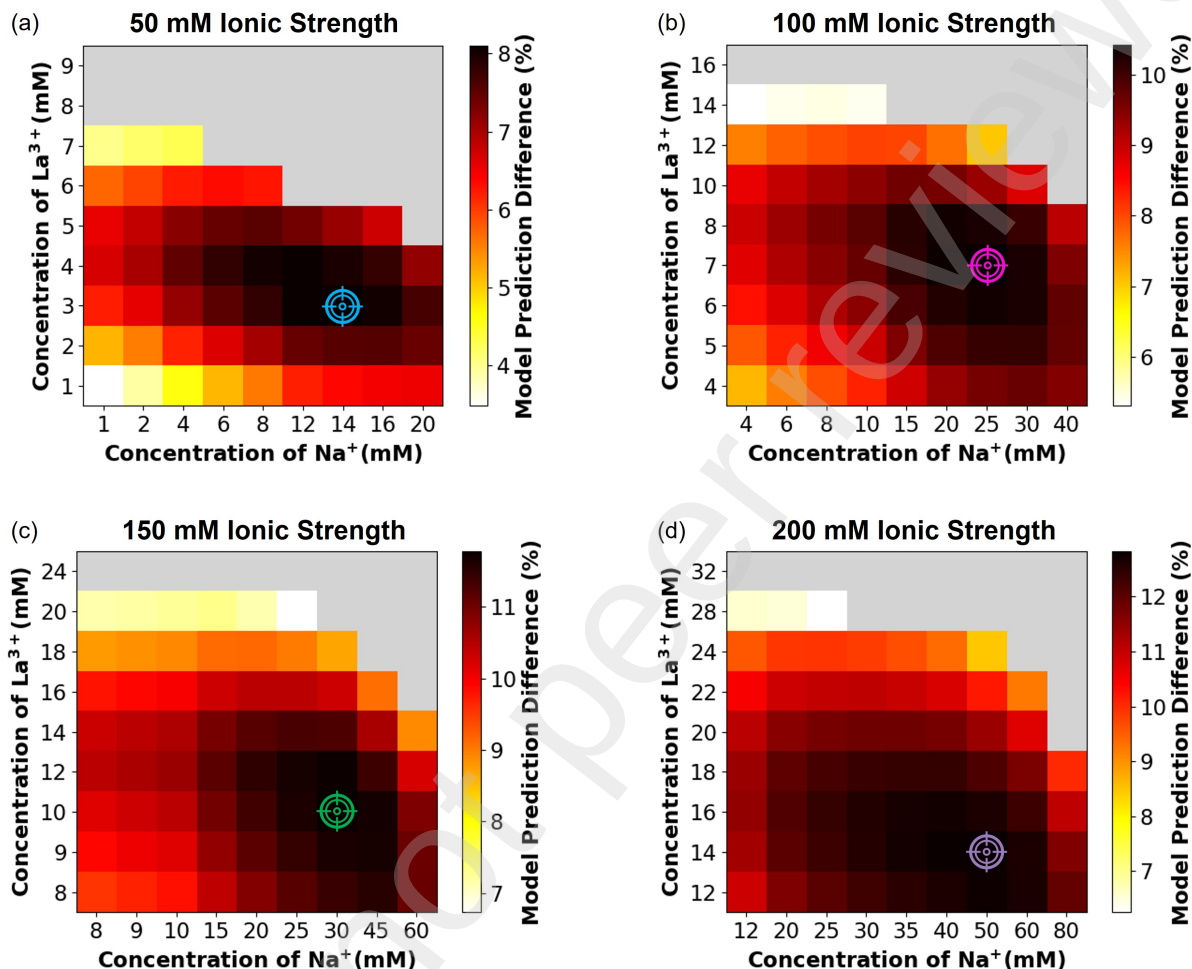


Figure 5: Guiding future NaCl-CaCl₂-LaCl₃ ternary experiments to discriminate between the Shedlovsky and MSA models at (a) 50 mM, (b) 100 mM, (c) 150 mM, and (d) 200 mM ionic strengths. The dart graphics points to the optimal (i.e., recommended) experiments to produce the most difference in model predictions. The grey shaded regions indicate conditions under which a ternary system cannot be realized, as the required Ca²⁺ concentration (Figure S3) is either 0 mM or negative to satisfy the specified ionic strength.

Figure 5 and Figure S3 show that, at an ionic strength of 50 mM, the optimal experimental condition identified using the Hunter and Reiner [29] criterion corresponds to 14 mM Na⁺, 3 mM La³⁺, and 6 mM Ca²⁺ in the NaCl – CaCl₂ – LaCl₃ ternary system. At these concentrations, all three cations contribute appreciably to charge transport, as reflected by their transport numbers; Na⁺ : 0.15, Ca²⁺ : 0.15, and La³⁺ : 0.13, defined by Eq. (13b). The comparable magnitudes of these transport numbers indicate that the solution behaves as a

fully coupled ternary system, with each cation participating meaningfully in conductivity.

Under this ternary condition, the omission of explicit cation–cation interactions in the Shedlovsky model is expected to produce the largest deviation from both the experimental data and the MSA model predictions. Similar behavior is observed at the optimal conditions for ionic strengths of 100 (25 mM Na⁺, 7 mM La³⁺, and 11 mM Ca²⁺), 150 (30 mM Na⁺, 10 mM La³⁺, and 20 mM Ca²⁺), and 200 mM (50 mM Na⁺, 14 mM La³⁺, and 22 mM Ca²⁺), where all cations continue to exhibit significant contributions to charge transport.

The optimal experimental conditions corresponding to these four ionic strengths were used to collect conductivity measurements to validate the model discrimination results. Figure 6 and Table 5 show that experimental validation under these optimal design conditions demonstrates strong agreement with the MSA model and systematic overprediction by the Shedlovsky model. This observed behavior is consistent with the hypothesis that the Shedlovsky model may yield larger prediction errors in systems with a higher overall cationic charge because it does not explicitly account for cation-cation interactions. Collectively, these results indicate that the MSA model provides a more appropriate predictive framework for salt systems containing more than two dissolved salts.

Model	MSE ($\mu\text{S} \cdot \text{cm}^{-1}$) ²	MAE ($\mu\text{S} \cdot \text{cm}^{-1}$)	MAPE (%)
Shedlovsky	1,095,630	850	8.8
MSA	123,375	295	4.4

Table 5: Error metrics for the conductivity predictions in $\mu\text{S} \cdot \text{cm}^{-1}$ of the Shedlovsky and MSA models in NaCl – CaCl₂ – LaCl₃ ternary system representative of multicomponent electronic waste, where the lowest values are bold.

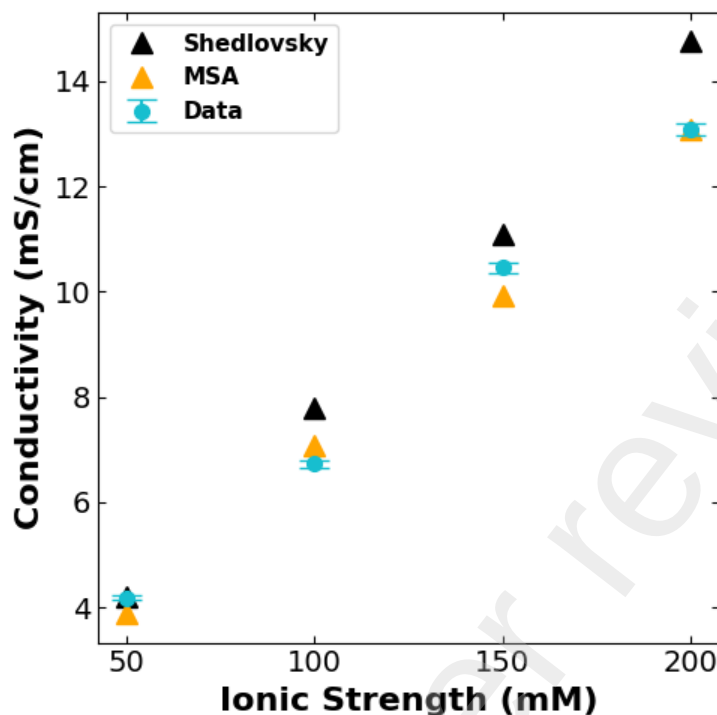


Figure 6: Conductivity predictions of the Shedlovsky and MSA models at the optimal experimental conditions corresponding to four ionic strengths in the $\text{NaCl} - \text{CaCl}_2 - \text{LaCl}_3$ ternary system, as identified using the Hunter and Reiner [29] model discrimination criterion. At each ionic strength (50, 100, 150, and 200 mM), the corresponding concentrations (in mM) for Na^+ , La^{3+} , and Ca^{2+} are: (14, 3, 6), (25, 7, 11), (30, 10, 20), and (50, 14, 22), respectively.

4.4. The soft sensors estimate of ion concentrations in membrane permeate solution validate ICP-OES data

Sections 4.1, 4.2, and S4 demonstrate that the Shedlovsky and MSA models accurately predict experimental conductivity measurements for a range of feed solutions. Building on these results, this section applies a soft sensor framework that incorporates the Shedlovsky and MSA models to estimate real-time ion concentrations from conductivity measurements obtained during diafiltration experiments.

As described in Section 3.2, the feed and diafiltrate introduced into the diafiltration apparatus consist of acidic CoCl_2 solutions with concentrations of 1 mM and 50 mM, respectively, maintained at pH 3 by adding the corresponding concentration of HCl. Figure 7 shows the conductivity measurements of these solutions at 25°C obtained from batch experiments. By treating the solution as a binary mixture of the primary salt (CoCl_2) and HCl, the Shedlovsky and MSA-based soft sensors accurately reproduce the conductivity of the feed and diafiltrate solutions, yielding MAPE values of 3.9 and 3.4%, respectively. These

results confirm that the soft sensor framework can reliably capture the conductivity behavior of the feed solutions used in the diafiltration experiment.

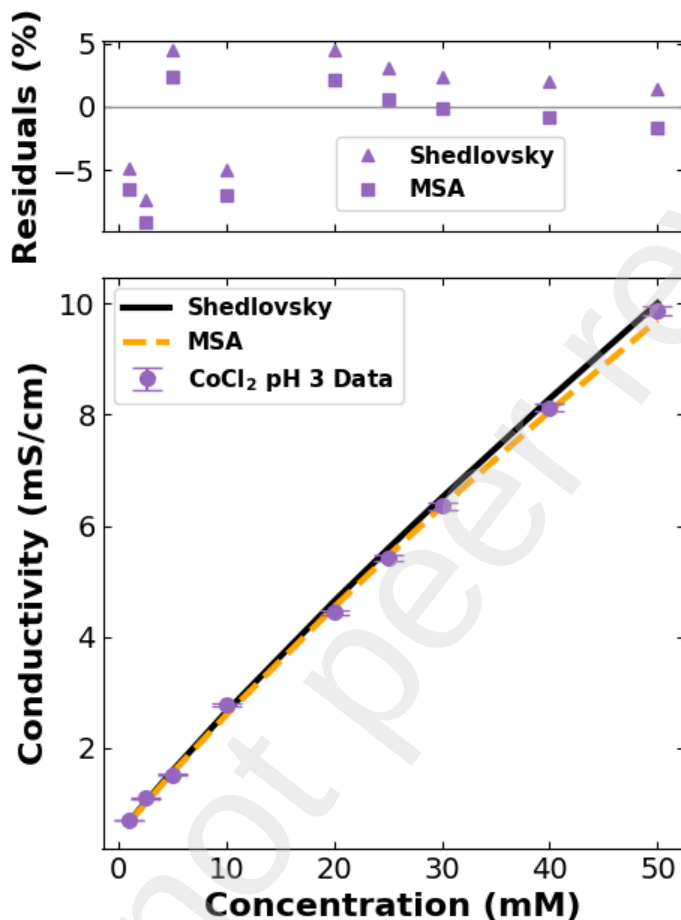


Figure 7: Shedlovsky and MSA models prediction for the conductivity of the acidic CoCl_2 solution used in the feed and diafiltrate of the diafiltration apparatus (Figure 2). The top panel presents the percent errors of the model predictions.

During the diafiltration experiment, diafiltrate is continuously introduced on the upstream side of the membrane, while the conductivity and pH of the permeate outlet are monitored over time, as shown in Figure 8. Figure 8 illustrates the temporal evolution of the permeate pH during the diafiltration experiment. The observed changes in the pH of the permeate reflect the combined effects of the addition of diafiltrate to the feed and the transport of ionic species, including H^+ , across the membrane. As the composition of the feed evolves during the experiment, the partitioning and transport of ions through the mem-

brane alter the acidity of the permeate solution. Given the potential for pH excursions to alter the total conductivity of the CoCl_2 solution through the mobility of H^+ , it is critical to quantify the specific impact of these pH fluctuations on the observed conductivity.

4.4.1. MSA model simulations elucidate the effect of pH on conductivity

The effect of pH on conductivity was investigated using MSA simulations of a simple single salt (LiCl) system at fixed pH values of 2, 3, 4, and 7 over a concentration range of 1 - 50 mM (Figure S4).

Figure S4 shows that decreasing the pH from neutral to 4 does not have a significant effect on the conductivity of the salt solution, due to the low concentration of H^+ , causing the solution to exhibit a single salt behavior. However, at pH values below 4, the conductivity increases significantly due to the pronounced presence of H^+ . For example, at pH 3, the conductivity of the salt solution increases by $\sim 65\%$ compared to its value at pH 7, while at lower pH values, such as pH 2, the conductivity increases by $\sim 638\%$ (see Figure S4). This trend is driven by the increasing concentration of H^+ and because H^+ has a diffusion coefficient and limiting molar conductivity nine times greater than those of other cations (Li^+), H^+ exhibits a higher transport number (e.g., $\text{H}^+ : 0.35$, $\text{Li}^+ : 0.19$ at pH 2).

In aqueous solutions, H^+ exhibits a higher diffusion coefficient relative to other cations because its migration does not rely solely on the physical movement of its mass through the solvent. Instead, H^+ transport occurs primarily via the Grothuss mechanism, a proton hopping process in which the electrical charge is transferred along a hydrogen-bonded network of water molecules rather than through the movement of an individual H^+ ion [57, 58]. A proton (i.e., H^+) associates itself with a water molecule to form hydronium ion, which again does not migrate across the solvent but transfers the extra proton to a neighboring water molecule through the hydrogen bond [57, 58]. This continuous proton hopping transfers charge rapidly.

The trends observed for the LiCl system establish a baseline for the pH-dependent behavior of solution conductivity. Consequently, to ensure an accurate estimation of the concentration of Co^{2+} in the permeate outlet during diafiltration, the true conductivity of CoCl_2 must be deconvoluted from the permeate conductivity measurements.

4.4.2. MSA model deconvolutes the conductivity of CoCl_2 from permeate conductivity measurements obtained during diafiltration

The CoCl_2 conductivity was isolated from the permeate conductivity measurements by modeling the acidic CoCl_2 solution as a binary mixture of CoCl_2 and HCl . This modeling

approach was necessitated by the observed pH range (2.6-3.3), as shown in Figure 8, a regime in which H^+ concentrations significantly affect the conductivity of the solution. For each data point, the concentration of H^+ was determined from the measured pH as follows:

$$pH = -\log_{10} c_{H^+} \implies c_{H^+} = 10^{-pH} \quad (24)$$

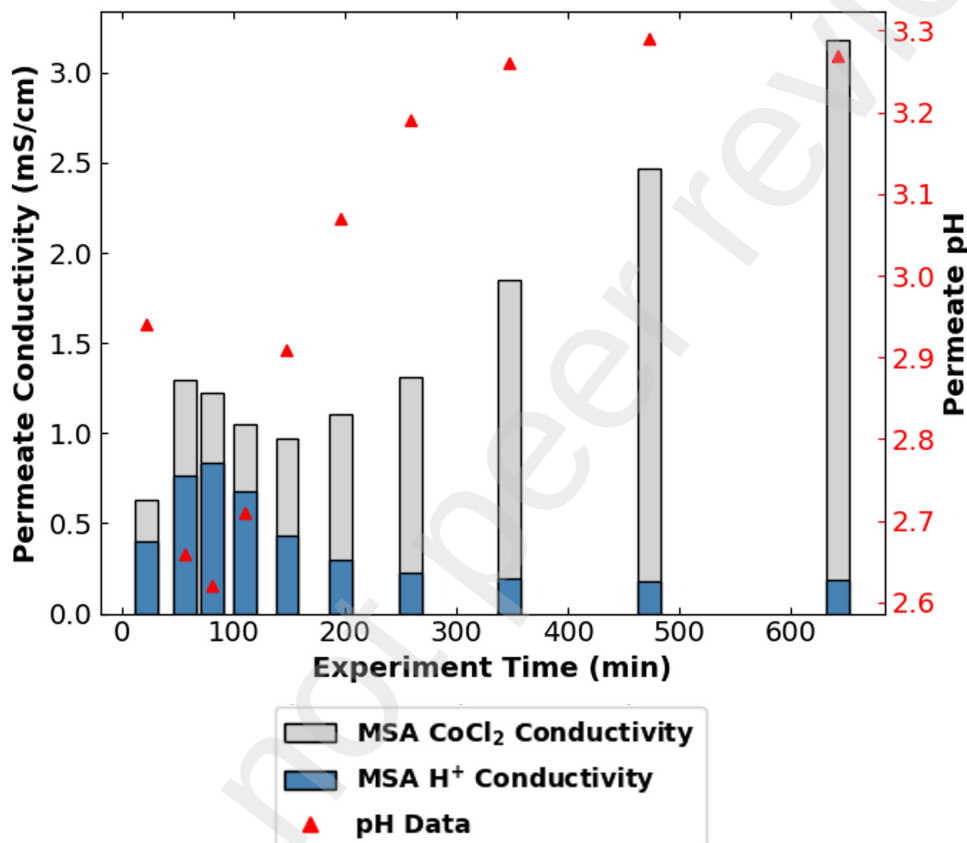


Figure 8: MSA-based isolation of $CoCl_2$ conductivity (grey-shaded regions) from permeate conductivity measurements (bars) during diafiltration. The blue-shaded regions indicate the calculated H^+ conductivity.

The calculated H^+ concentration was used to compute the corresponding H^+ conductivity using the MSA model. The $CoCl_2$ conductivity was obtained as the difference between the measured permeate conductivity and the calculated H^+ conductivity. Figure 8 compares this isolated $CoCl_2$ conductivity with the measured permeate conductivity. As shown in Figure 8, at experimental times of 22, 56, 81, and 111 minutes, the concentration of Co^{2+} in the permeate remains notably low, while the pH ranges between 2.6 and 2.9. Under these conditions, H^+ ions contribute more than 50% of the measured conductivity per the

MSA model. This dominance is attributed to both the higher concentration and mobility of H^+ relative to Co^{2+} , resulting in H^+ becoming the primary charge carrier in the permeate solution.

The decrease in pH observed between 22 and 111 minutes occurs because H^+ permeates through the membrane more readily than Co^{2+} . Because the permeate volume is initially small, even modest H^+ flux into the permeate leads to relatively high proton concentrations and lower pH values. In contrast, at experimental times beyond 111 minutes, as the permeate volume and the concentration of Co^{2+} increase, the pH rises above 2.9. Under these conditions, the contribution of H^+ to the measured conductivity of the permeate reduces to values below 40%, as charge transport becomes increasingly influenced by Co^{2+} ions.

Following successful deconvolution of the intrinsic CoCl_2 conductivity from permeate conductivity measurements, Shedlovsky and MSA-based soft sensors can be used to estimate Co^{2+} concentrations.

4.4.3. Ion concentration estimates from the soft sensors agree with ICP-OES measurements

The CoCl_2 conductivity in the permeate, obtained by removing the calculated conductivity of H^+ from the measured conductivity, is used to estimate Co^{2+} concentrations using the soft sensor frameworks. The soft sensor algorithm estimates the Co^{2+} concentration using a numerical inversion technique, identifying the concentration that produces zero residuals between the CoCl_2 conductivity and the values predicted by the Shedlovsky and MSA models.

Figure 9 compares the estimated concentration of Co^{2+} from Shedlovsky and MSA-based soft sensors with data from ICP-OES analysis. Both soft sensors emulate ICP-OES measurements closely at experimental times beyond 81 minutes, when the flux of Co^{2+} through the membrane becomes significant and Co^{2+} concentration in the permeate outlet approaches or exceeds 1 mM. In contrast, at experimental times earlier than 81 minutes, both soft sensors overestimate the concentration of Co^{2+} . During this initial phase of the diafiltration, the flux of Co^{2+} through the membrane is minimal, resulting in dilute permeate concentrations, typically below 1 mM. Concurrently, the H^+ concentration is relatively high. Because H^+ has a molar conductivity significantly higher than Co^{2+} , even small variations in pH lead to disproportionately large changes in its modeled conductivity contribution. This sensitivity highlights the importance of characterizing measurement uncertainty, as small errors in pH sensing can propagate into significant variances in the predicted concentrations. Consequently, during these early time intervals at low pH, the inversion of the conductivity data temporarily reallocates a portion of the measured conductivity to Co^{2+} , leading to an

overestimation of its concentration.

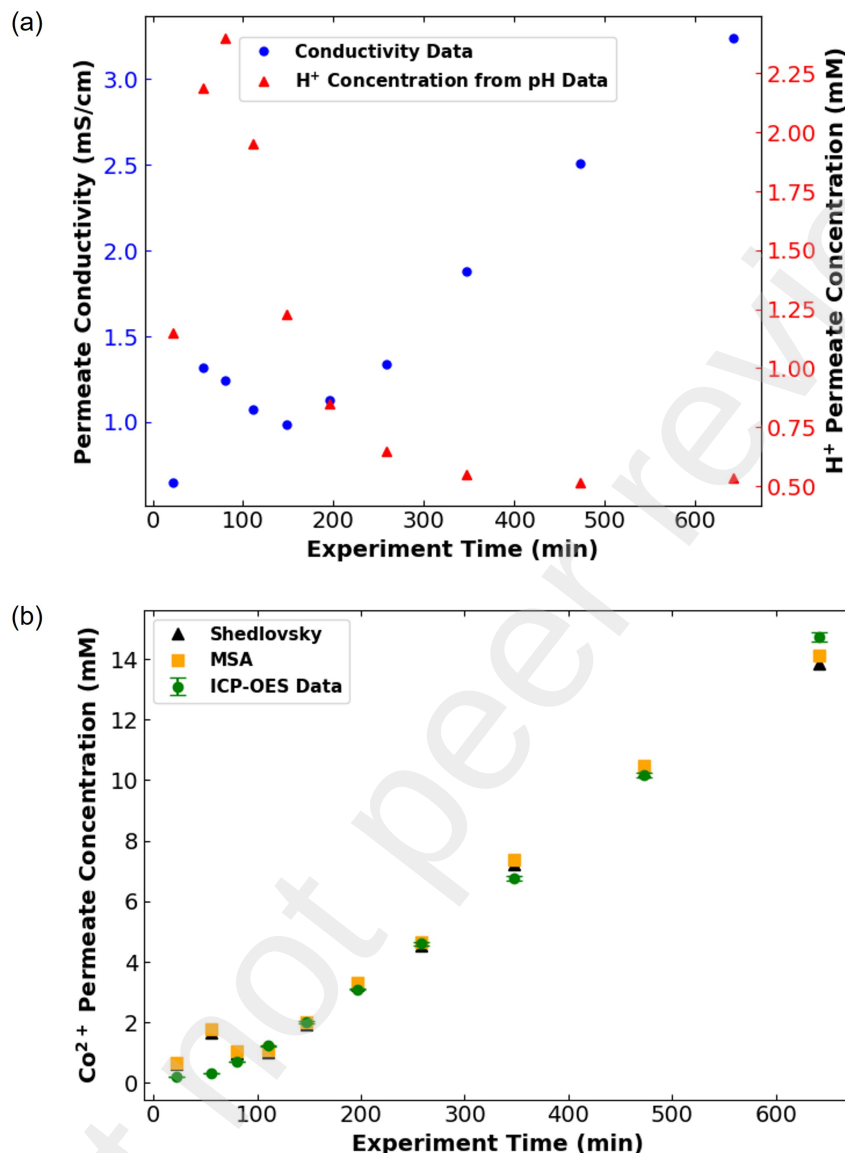


Figure 9: (a) Measured H⁺ concentration and conductivity of the acidic CoCl₂ permeate solution. (b) Co²⁺ concentration in the permeate outlet predicted using Shedlovsky and MSA-based soft sensors.

5. Conclusions

This study extends the original Shedlovsky formulation from single salt systems to binary and ternary salt mixtures. We demonstrate that while the Shedlovsky model accurately predicts conductivity measurements in single salt solutions, its predictive capability diminishes in binary and ternary salt mixtures due to the omission of cation-cation interactions.

Conversely, the MSA model exhibits superior performance in these complex multi-cation systems by explicitly accounting for all ionic interactions. A key contribution of this work is the elucidation of the impact of pH on the conductivity of acidic solutions, such as lithium-ion battery leachate. Sensitivity analysis using the MSA framework reveals that while the contribution of H^+ ions is negligible between pH 4 and 7, it becomes a dominant factor below pH 4 due to the higher limiting molar conductivity and diffusion coefficient of protons relative to other cations. By modeling the pH control agent (e.g., HCl) as a participating salt, we demonstrated that both Shedlovsky and MSA-based soft sensors can provide reliable estimates of cation concentrations from conductivity measurements in complex ionic environments, such as those encountered in membrane-based separations (e.g., recovery of cobalt ions from acidic cobalt chloride). Ultimately, these findings suggest that coupling conductivity sensors with rigorous physics-based conductivity models, together with complementary membrane transport models (e.g., Ouimet et al. [9, 46]), within a soft sensor framework provides a rapid, efficient, and cost-effective approach for real-time monitoring of ion concentrations in complex aqueous systems.

While the proposed framework performs well across a wide range of aqueous conditions, limitations remain in highly acidic, dilute environments. At pH levels below 3 and cation concentrations less than 1 mM, both Shedlovsky and MSA-based soft sensors tend to overestimate cation concentrations, driven by the disproportionately high transport number of H^+ ions. To address this limitation, future work should incorporate additional physicochemical effects into the conductivity frameworks, including ion pairing (or speciation) mechanisms and solvent dielectric decrement, both of which are likely to play significant roles in highly acidic systems containing dilute concentrations of multivalent ions.

6. Author Contributions

Shammah Lilonfe: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing– original draft, Writing– review & editing. **Laurianne Estrada:** Formal analysis, Data curation, Writing– original draft, Writing– review & editing. **Pratham Singh:** Formal analysis, Data curation, Writing– original draft, Writing– review & editing. **Jonathan A. Ouimet:** Formal analysis, Data curation, Writing– review & editing. **William A. Phillip:** Supervision, Project administration, Funding acquisition, Conceptualization, Methodology, Formal analysis, Data curation, Writing– review & editing. **Alexander W. Dowling:** Supervision, Project administration, Funding acquisition, Conceptualization, Methodology, Software, Formal analysis, Investigation, Writ-

ing— review & editing.

7. Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information: Soft Sensors Enable Real-Time Ion Concentration Measurements

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S1. Capturing conductivity sensor calibration campaign using NIST standards and the Shedlovsky and MSA models

Figure S1 illustrates the instrument bias in the experimental conductivity measurements from the uncalibrated Probe 1 and Probe 2, benchmarked against NIST (KCl) conductivity standard and the predictions of the Shedlovsky and MSA models. Probe 1 and Probe 2 show -13 and -11% instrument bias, respectively.

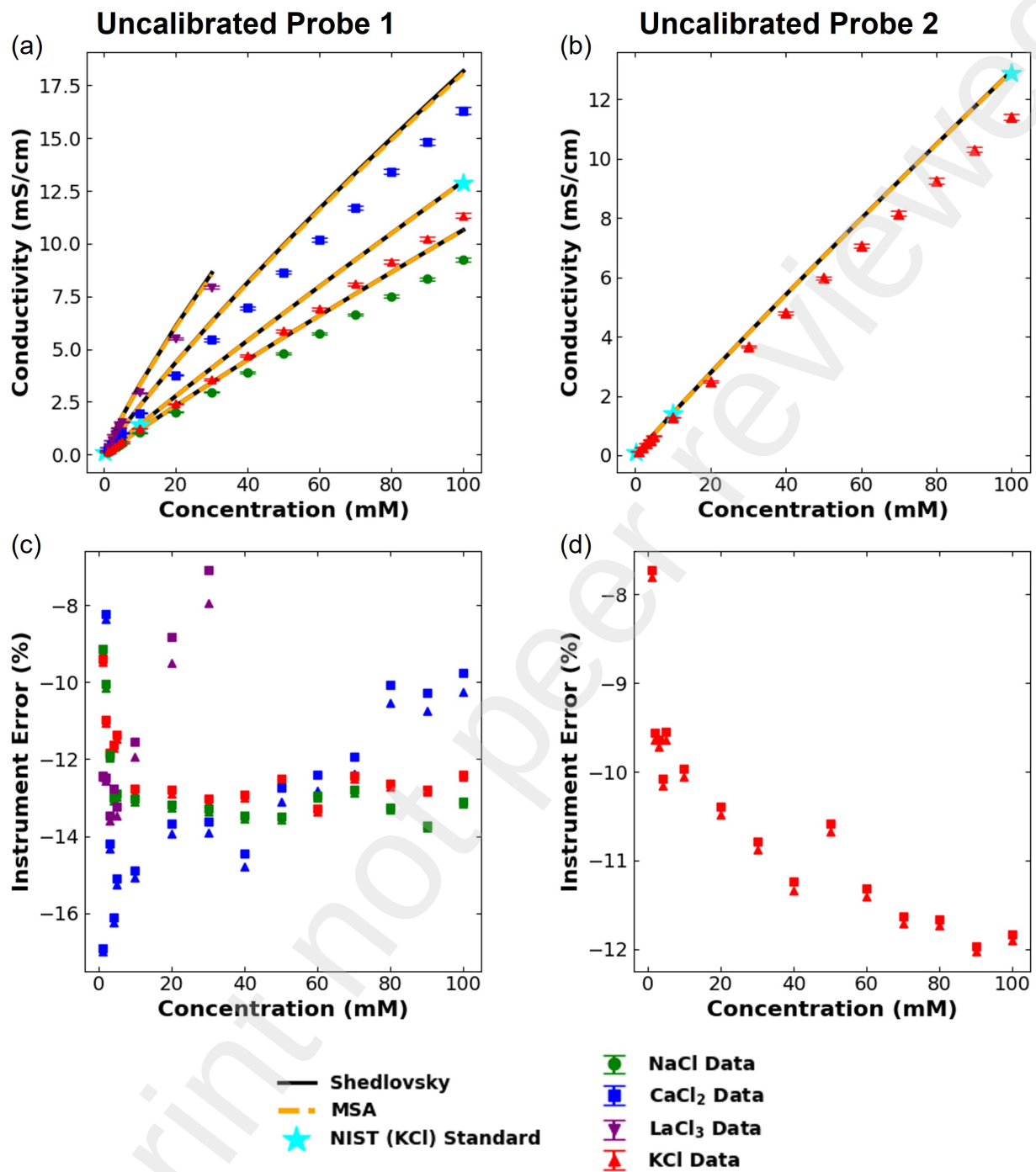


Figure S1: (a) and (b) compares the conductivity data from uncalibrated Probe 1 and Probe 2 with NIST (KCl) standards and the Shedlovsky and MSA models. (c) and (d) show the instrument bias of Probe 1 and Probe 2, respectively.

S2. Calibration of conductivity sensors using NIST standards and linear regression

To ensure accurate conductivity measurements in the ionic feed solutions, the conductivity sensor was calibrated using NIST standards by regressing the cell constant Φ from the standard conductivity data according to Eq. (3). Eq. (3) can be represented in the following mathematical form:

$$\mathbf{y} = \Phi \mathbf{x} \quad (\text{S1})$$

where, $\mathbf{x} \in \mathbb{R}^3$ are the conductance ($\frac{I}{V}$) in S and $\mathbf{y} \in \mathbb{R}^3$ are the conductivity in $\text{S} \cdot \text{cm}^{-1}$. The cell constant Φ is estimated using the linear least squares:

$$\hat{\Phi} = (\mathbf{x}^T \mathbf{x})^{-1} \mathbf{x}^T \mathbf{y} \quad (\text{S2})$$

The uncertainty σ_{Φ} associated with the regressed cell constant $\hat{\Phi}$ is calculated using the first-order uncertainty propagation equation:

$$\begin{aligned} \sigma_{\Phi}^2 = & \left(\frac{\partial \Phi}{\partial \chi} \sigma_{\chi} \right)^2 + \left(\frac{\partial \Phi}{\partial V} \sigma_V \right)^2 + \left(\frac{\partial \Phi}{\partial I} \sigma_I \right)^2 \\ & + 2 \frac{\partial \Phi}{\partial \chi} \frac{\partial \Phi}{\partial V} \text{cov}(\chi, V) + 2 \frac{\partial \Phi}{\partial \chi} \frac{\partial \Phi}{\partial I} \text{cov}(\chi, I) + 2 \frac{\partial \Phi}{\partial V} \frac{\partial \Phi}{\partial I} \text{cov}(V, I) \end{aligned} \quad (\text{S3})$$

where, σ_{χ} , σ_I , and σ_V are the measurement errors of the conductivity, current, and voltage measurements, respectively. The values of σ_{χ} , σ_I , and σ_V are 0.93%, 1.00%, and 0.14% of the measurements, respectively, as presented in Section 3.3.

Assuming insignificant covariance between the measurements, Eq. (S3) becomes:

$$\sigma_{\Phi}^2 \approx \left(\frac{\partial \Phi}{\partial \chi} \sigma_{\chi} \right)^2 + \left(\frac{\partial \Phi}{\partial V} \sigma_V \right)^2 + \left(\frac{\partial \Phi}{\partial I} \sigma_I \right)^2 \quad (\text{S4})$$

Substituting $\frac{\partial \Phi}{\partial \chi} = \frac{V}{I}$, $\frac{\partial \Phi}{\partial V} = \frac{\chi}{I}$, and $\frac{\partial \Phi}{\partial I} = -\frac{\chi V}{I^2}$ into Eq. (S4) yields:

$$\sigma_{\Phi} = \sqrt{\left(\frac{V}{I} \sigma_{\chi} \right)^2 + \left(\frac{\chi}{I} \sigma_V \right)^2 + \left(\frac{\chi V}{I^2} \sigma_I \right)^2} \quad (\text{S5})$$

S3. Regressed cell constants for Probe 1 and Probe 2

Table S1 presents the cell constants of Probe 1 and Probe 2 (with their associated uncertainty), determined by regression against the NIST (KCl) conductivity standard using

Eq. (S2) and Eq. (S5).

	Manufacturer's cell constant (cm^{-1})	Regressed cell constant (cm^{-1})	Calculated uncertainty (cm^{-1})
Probe 1	0.43	0.497	0.010
Probe 2	0.43	0.486	0.010

Table S1: Regressed cell constants for Probe 1 and Probe 2 from NIST (KCl) standards.

S4. Shedlovsky and MSA models accurately predict the conductivity of single salt solutions containing ions typical of lithium-ion battery leachate

Figure S2 presents the conductivity predictions of the Shedlovsky and MSA models for single salt LiCl, CoCl_2 , and HCl solutions prepared using a procedure similar to that described in Section 3.1. As shown in Figure S2 and summarized in Table S2, both models accurately reproduce experimental conductivity measurements, with MAPE values below 6%. Compared to the single salt solution of electronic waste analyzed in Section 4.1, the MSA model performs comparatively better than the Shedlovsky model in the lithium-ion battery leachate system due to the smaller size of the cations. In the presence of an external electric field, small ions exhibit reduced electrophoretic effects, resulting in lower hydrodynamic drag on neighboring ions [22, 56].

Model	MSE ($\mu\text{S} \cdot \text{cm}^{-1}$) ²	MAE ($\mu\text{S} \cdot \text{cm}^{-1}$)	MAPE (%)
HCl			
Shedlovsky	236,863	318	4.5
MSA	326,098	346	4.4
LiCl			
Shedlovsky	17,048	104	5.9
MSA	10,297	84	5.3
CoCl_2			
Shedlovsky	24,742	132	4.5
MSA	9,763	81	3.5

Table S2: Error metrics for the conductivity predictions in $\mu\text{S} \cdot \text{cm}^{-1}$ of the Shedlovsky and MSA models in single salt solutions containing ions typical of lithium-ion battery leachate, where the lowest values are bold.

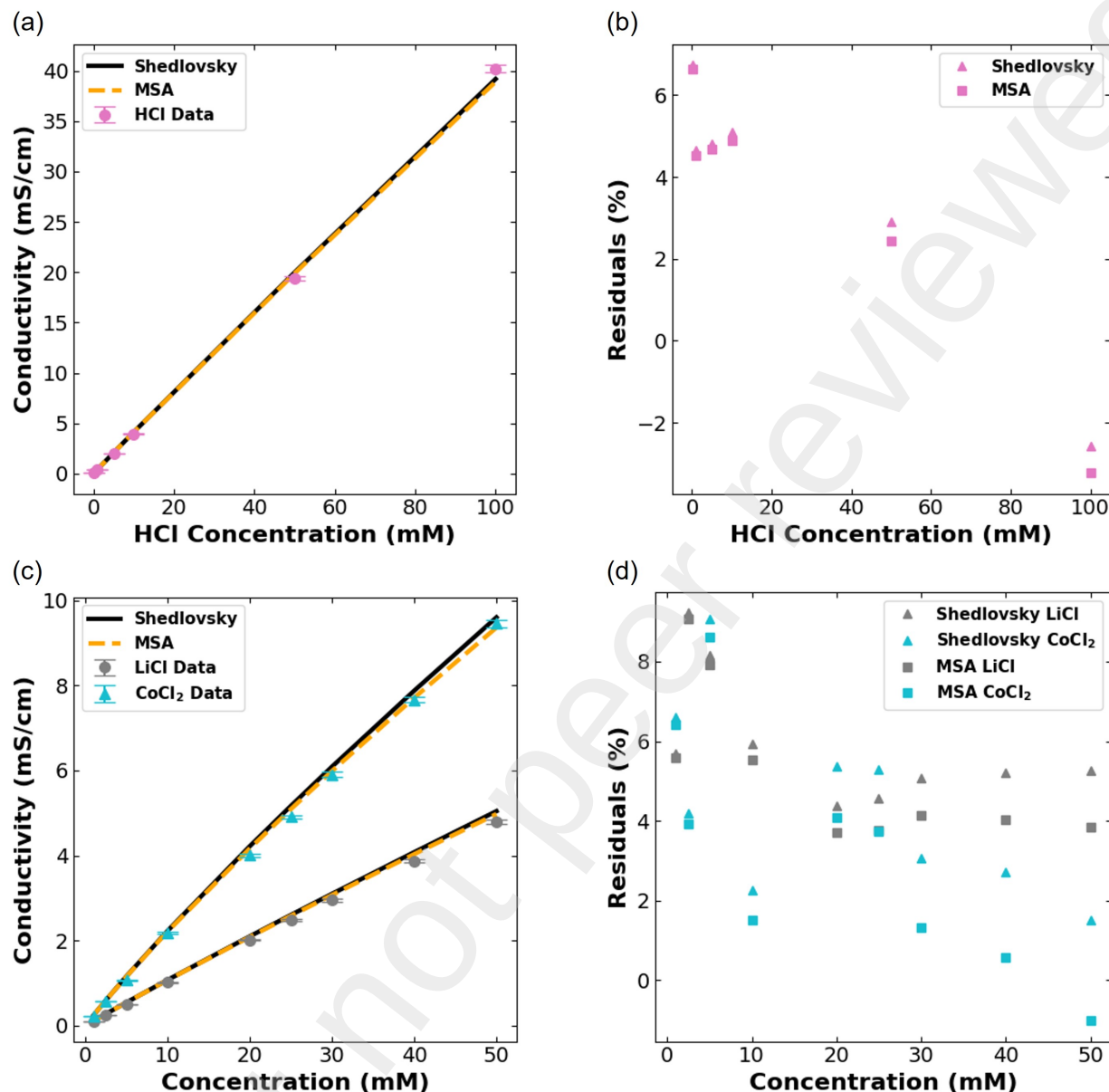


Figure S2: Shedlovsky and MSA models conductivity predictions for (a) HCl and (c) LiCl and CoCl₂ single salt solutions commonly found in lithium-ion battery leachate. The percent error of the conductivity predictions for HCl, LiCl, and CoCl₂ are shown in (b) and (d), respectively.

S5. Concentrations of Ca²⁺ corresponding to selected Na⁺ and La³⁺ concentrations at the specified ionic strengths in the ternary system

Figure S3 shows the concentration of Ca²⁺ corresponding to various Na⁺ and La³⁺ concentrations at selected ionic strengths in the NaCl-CaCl₂-LaCl₃ ternary system, used to discriminate between the Shedlovsky and MSA models.

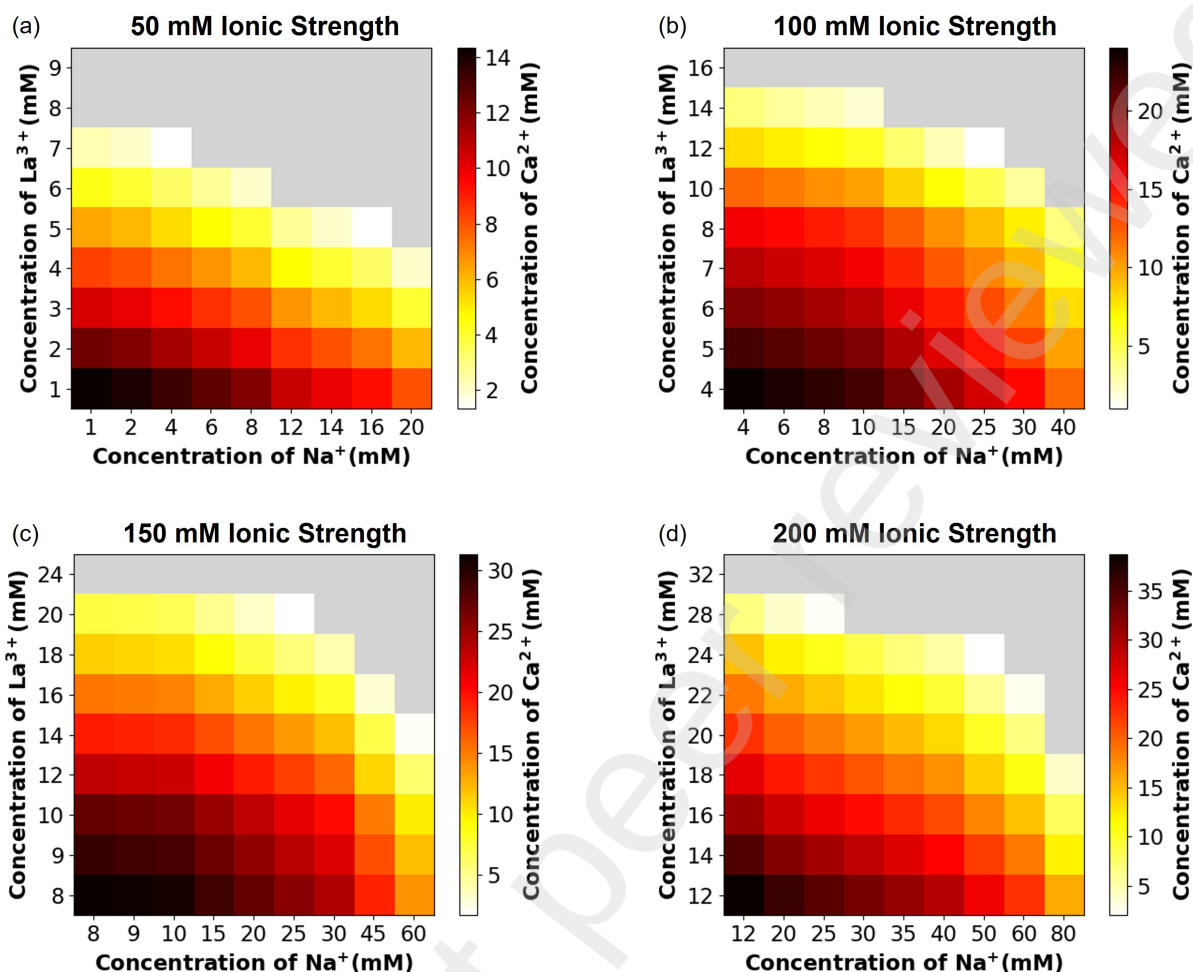


Figure S3: Calculated concentration of Ca^{2+} corresponding to selected Na^{+} and La^{3+} concentrations at ionic strengths of (a) 50 mM, (b) 100 mM, (c) 150 mM, and (d) 200 mM in the ternary salt system for model discrimination between the Shedlovsky and MSA frameworks. The grey shaded regions indicate conditions under which a ternary system cannot be realized, as the required Ca^{2+} concentration is either 0 mM or negative to satisfy the specified ionic strength.

S6. The effect of pH on the conductivity measurements of ionic solutions

The effect of pH on the conductivity of ionic solutions was studied by evaluating the MSA model at 25°C and various concentrations of LiCl solution (with water as solvent) given a desired constant pH level, as shown in Figure S4.

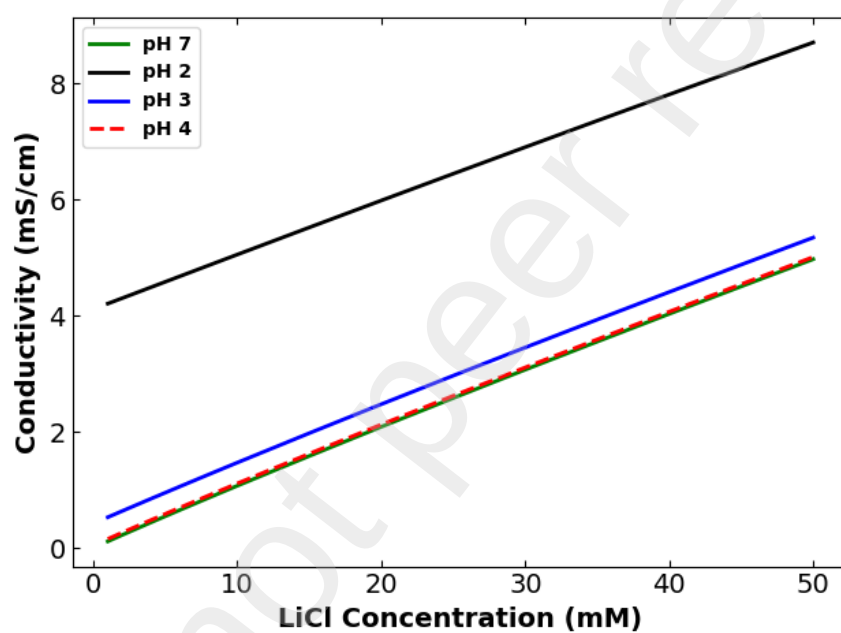


Figure S4: Conductivity of LiCl solution at different constant pH conditions.